The Fundamental Theorem of Calculus

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: evaluating definite integrals

Evaluate each definite integral:

a $\int_1^3 (2x+1)\,dx$

b $\int_0^{\pi} \sin x\,dx$

c $\int_1^4 \sqrt{x}\,dx$

2 [Foundational] • Target: evaluating definite integrals

Evaluate $\int_0^1 e^{2x}\,dx$, to three decimal places.

3 [Foundational] • Target: differentiating an integral

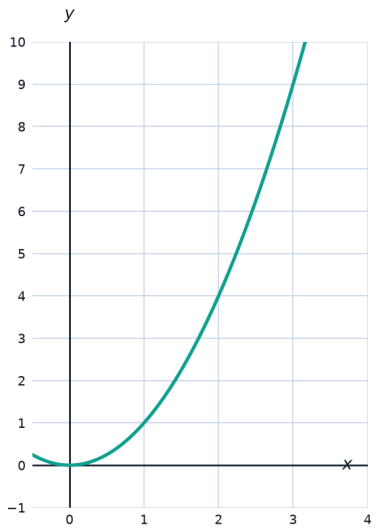
Use the Fundamental Theorem, Part 1, to find $\frac{d}{dx}\int_2^x (t^3+1)\,dt$.

4 [Foundational] • Target: differentiating an integral

Find $\frac{d}{dx}\int_1^{x^2} \sin(t)\,dt$ using the chain rule with FTC.

5 [Foundational] • Target: Average value and applications

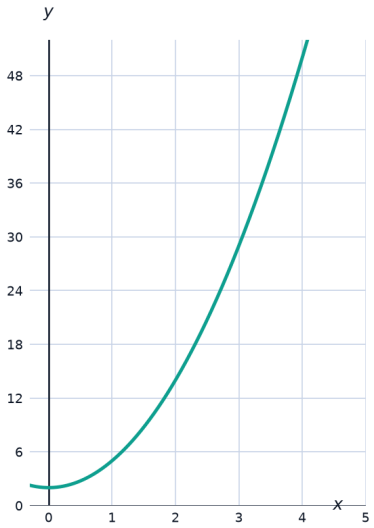
Find the average value of $f(x) = x^2$ on $[0, 3]$.



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6 [Foundational] • Target: Average value and applications

A car's velocity is $v(t) = 3t^2 + 2$ m/s for $0 \le t \le 4$. Find the total distance traveled, and the average velocity over this interval.



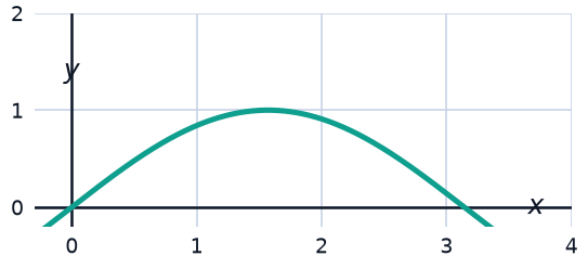
7 [Foundational] • Target: Further FTC practice

Find $\frac{d}{dx}\int_x^5 \cos(t^2)\,dt$, using the property $\int_a^b = -\int_b^a$ before applying FTC Part 1.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Further FTC practice

Find the average value of $f(x) = \sin x$ on $[0, \pi]$, to three decimal places.



9 [Intermediate] • Target: Mean Value Theorem for integrals

Find the value(s) of c guaranteed by the Mean Value Theorem for Integrals for $f(x) = x^2$ on $[0, 3]$ (where $f(c)$ equals the average value of f).

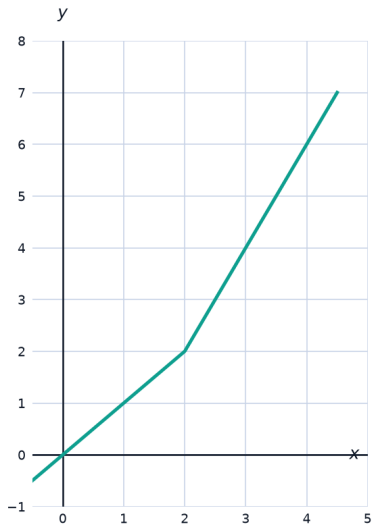
10 [Intermediate] • Target: Mean Value Theorem for integrals

Evaluate $\int_2^5 f(x)\,dx$ given that the average value of f on $[2, 5]$ is 7.

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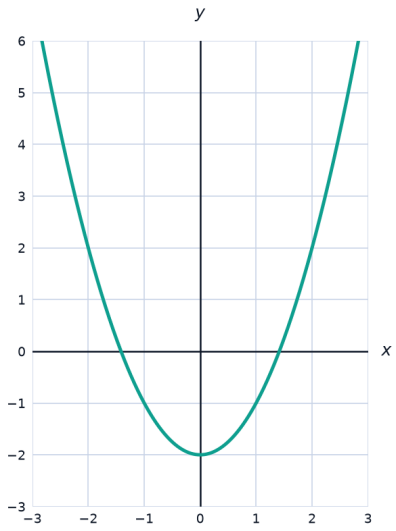
11 [Intermediate] • Target: Integrals of piecewise functions

Evaluate $\int_0^4 f(x)\,dx$ where $f(x) = x$ for $x \le 2$ and $f(x) = 2x - 2$ for $x > 2$.



12 [Intermediate] • Target: Visualizing the two parts of FTC together

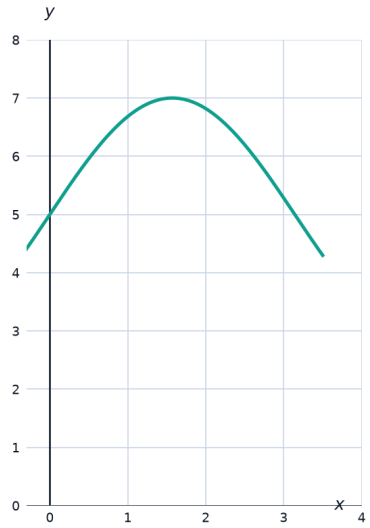
For $g(x) = \int_0^x (t^2 - 2)\,dt$, find $g'(x)$ using FTC Part 1, then find where g has a local minimum by analyzing the sign of $g'(x)$.



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13 **[Intermediate]** • **Target: A capstone FTC application**

The rate of water flowing into a tank is $r(t) = 5 + 2\sin(t)$ liters/minute. Find the total volume that flows in during $[0, \pi]$ minutes, using FTC.



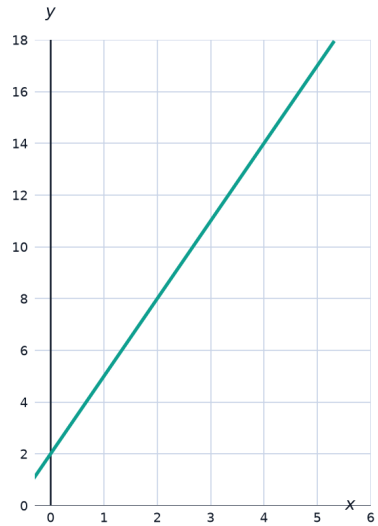
14 **[Intermediate]** • **Target: FTC Part 1 with more complex upper limits**

Find $\frac{d}{dx}\int_{x^2}^{5}\cos(t)\,dt$, using $\int_{x^2}^{5} = -\int_{5}^{x^2}$ before differentiating.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 **[Challenging]** • **Target: Net change and the FTC** • **Cross-Lesson: 14.01 Partial Derivatives and the Gradient**

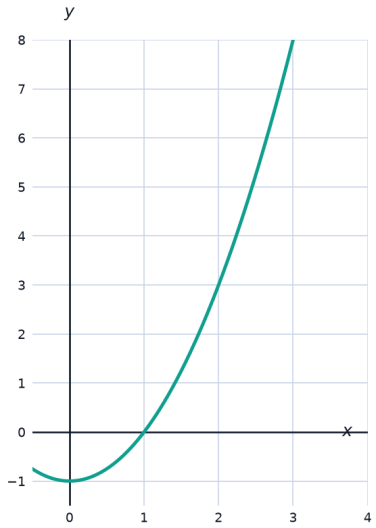
A tank starts with 200 liters of water. Water flows out at rate $r(t) = 3t + 2$ liters/minute for the first 5 minutes. Use FTC to find the amount of water in the tank at $t = 5$.



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16 **[Challenging]** • **Target: Visualizing FTC Part 2 as area** • **Cross-Lesson: 15.01 Double Integrals**

The graph shows $f(x) = x^2 - 1$ on $[0, 3]$. Use FTC to find $\int_0^3 (x^2 - 1)\,dx$ and explain why the result is not simply the area of the shaded region (part of the curve dips below the axis).



17 **[Challenging]** • **Target: Combining FTC Part 1 and Part 2 in one problem** • **Cross-Lesson: 16.01 Vector Fields and Line Integrals**

Let $F(x) = \int_1^x (3t^2 - 2)\,dt$. Find $F(2)$ using Part 2, then verify $F'(2)$ using Part 1 and confirm it matches direct substitution into the integrand.

18 **[Challenging]** • **Target: Integrals of piecewise functions** • **Cross-Lesson: 2.01 Evaluating Limits**

[Vector Calculus · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

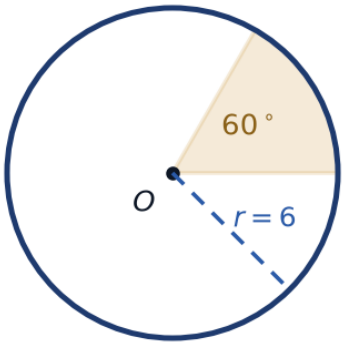
19 **[Challenging]** • **Target: Visualizing the two parts of FTC together** • **Cross-Lesson: 2.02 Continuity**

Level 3 · AdvancedEvaluate the definite integral using a substitution that transforms the limits: $\int_0^{\sqrt{\pi/2}} x\cos(x^2)e^{\sin(x^2)}\,dx$

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20 [Challenging] • Target: A capstone FTC application • Cross-Lesson: 3.01 The Derivative from First Principles

Use repeated integration by parts to evaluate the exact definite integral $\int_0^{\pi/2} x^2 \cos x \, dx$.



Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: evaluating definite integrals

FTC Part 2: $\int_a^b f(x)dx = F(b) - F(a)$ where $F' = f$. Evaluate $\int_0^{\pi} \cos x \, dx$:

a. 0

b. 2

c. -2

d. 1

22 [Foundational] • Target: differentiating an integral

If f is continuous on $[a, b]$ and F is any antiderivative, FTC Part 2 requires no assumption that F is:

a. equal to f

b. unique — any antiderivative works, since they differ by a constant

c. bounded on $[a, b]$

d. differentiable everywhere

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23 [Foundational] • Target: Average value and applications

Mean value theorem for integrals: $\int_a^b f\,dx = f(c)(b-a)$ for some $c \in (a,b)$. Requires f to be:

a. positive

b. integrable and bounded

c. continuous

d. analytic

24 [Foundational] • Target: Further FTC practice

Proof of FTC: $F(x) = \int_a^x f$. Key step: $F(x+h) - F(x) = \int_x^{x+h} f(t)dt \approx f(x) \cdot h$ by:

a. MVT for integrals

b. L'Hôpital

c. product rule

d. chain rule

25 [Foundational] • Target: Mean Value Theorem for integrals

FTC Part 1 is proved by constructing:

a. any antiderivative

b. $G(x) = \int_a^x f\,dt$ and showing $G' = f$

c. Taylor series

d. Riemann sums

26 [Foundational] • Target: Integrals of piecewise functions

Net Change Theorem states $\int_a^b f'\,dx = f(b) - f(a)$. This is:

a. MVT

b. FTC Part 2

c. EVT

d. Rolle's Theorem

27 [Foundational] • Target: Visualizing the two parts of FTC together

FTC Part 1 shows every continuous function has an:

a. antiderivative

b. derivative everywhere

c. limit

d. inflection point

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28 [Foundational] • Target: A capstone FTC application

A particle has velocity $v(t) = 2t$ m/s. Find its net displacement over $[1, 4]$.

a. 17

b. 15

c. 13

d. -15

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: FTC Part 1 with more complex upper limits

Find the area of the region bounded by $y = 3x$ and $y = x^2$ in the first quadrant.

a. $\frac{27}{3}$

b. $\frac{27}{6}$

c. $\frac{27}{2}$

d. $\frac{33}{6}$

30 [Intermediate] • Target: Net change and the FTC

Evaluate the definite integral of the constant function: $\int_8^{11} 21\,dx$.

a. 21

b. 84

c. 42

d. 63

31 [Intermediate] • Target: Visualizing FTC Part 2 as area

FTC allows computing $\int_0^1 x^{100}\,dx$ via:

a. L'Hôpital

b. power rule antiderivative

c. IBP

d. squeeze theorem

32 [Intermediate] • Target: Combining FTC Part 1 and Part 2 in one problem

Average value of f on $[a, b]$:

a. $\frac{f(a)+f(b)}{2}$

b. $\frac{1}{b-a}\int_a^b f\,dx$

c. $\int_a^b f\,dx$

d. $f\left(\frac{a+b}{2}\right)$

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33 **[Intermediate]** • **Target: evaluating definite integrals**

Area under one arch of $y = \sin x$ (from 0 to π):

a. 0

b. 2

c. 1

d. π

34 **[Intermediate]** • **Target: differentiating an integral**

Taylor integral remainder: $R_n = \int_a^x \frac{(x-t)^{n-1}}{(n-1)!} f^{(n)}(t)\,dt$. Requires $f \in$

a. C^0

b. C^n

c. C^∞

d. Lipschitz

35 **[Intermediate]** • **Target: Average value and applications**

Total area between $y = x$ and x-axis on $[-2, 2]$:

a. 0

b. 2

c. 4

d. 8

36 **[Intermediate]** • **Target: Further FTC practice**

Find the value of c guaranteed by the Mean Value Theorem for $f(x) = \frac{1}{x}$ on $[1, 16]$.

a. $\frac{17}{2}$

b. 8

c. 4

d. $\frac{19}{2}$

37 **[Intermediate]** • **Target: Mean Value Theorem for integrals**

$v(t) = 3t^2 - 2t$ on $[0, 2]$. Net displacement:

a. 2

b. 4

c. 6

d. 8

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

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38 [Challenging] • Target: Integrals of piecewise functions • Cross-Lesson: 2.01 Evaluating Limits

Net change: if velocity $v(t) = 3t^2$, displacement from $t = 0$ to $t = 2$ is

a. 8

b. 12

c. 6

d. 4

39 [Challenging] • Target: Visualizing the two parts of FTC together • Cross-Lesson: 2.02 Continuity

Logarithmic integral $Li(x) = \int_2^x \frac{dt}{\ln t} \Rightarrow Li'(x) =$

a. $x/\ln x$

b. $1/\ln x$

c. $\ln x$

d. $1/(x\ln x)$

40 [Challenging] • Target: A capstone FTC application • Cross-Lesson: 3.01 The Derivative from First Principles

Use Green's theorem to find the area enclosed by the astroid $x = 1\cos^3 t,\ y = 1\sin^3 t$.

a. π

b. $\frac{3\pi}{4}$

c. $\frac{3\pi}{8}$

d. $\frac{3\pi}{16}$

41 [Challenging] • Target: FTC Part 1 with more complex upper limits • Cross-Lesson: 3.02 Differentiation Rules: Power, Product, Quotient, Chain

Stokes' theorem $\iint_S (abla \times \mathbf{F}) \cdot d\mathbf{S} = \oint_C \mathbf{F} \cdot d\mathbf{r}$ generalises FTC to:

a. $\mathbb{R}^3$ surfaces

b. scalar fields

c. differential forms on manifolds

d. $\mathbb{R}^2$ only

42 [Challenging] • Target: Net change and the FTC • Cross-Lesson: 4.01 Antiderivatives and Indefinite Integrals

For f continuous on $[a, b]$, the function $F(x) = \int_a^x f(t)dt$ is guaranteed to be:

a. differentiable on (a, b), with $F \doteq f$

b. only continuous, not differentiable

c. a polynomial

d. constant

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43 [Challenging] • Target: Visualizing FTC Part 2 as area • Cross-Lesson: 4.02 Riemann Sums and Definite Integrals

By the Mean Value Theorem, find $c \in (0, 4)$ with $f'(c) = \frac{f(4) - f(0)}{4 - (0)}$ for $f(x) = x^2$.

a. 3

b. 1

c. 0

d. 2

44 [Challenging] • Target: Combining FTC Part 1 and Part 2 in one problem • Cross-Lesson: 6.01 Related Rates

Green's theorem area formula: $A = \frac{1}{2}\oint_C (x\,dy - y\,dx)$. Comes from $Q_x - P_y =$

a. 0

b. 1

c. 2

d. $x + y$

45 [Challenging] • Target: evaluating definite integrals • Cross-Lesson: 6.02 Optimization and Curve Sketching

MVT for integrals: $f(x) = x^2$ on $[0, 3]$. Value of c:

a. $\sqrt{3}$

b. $3/\sqrt{3}$

c. $3/2$

d. $\sqrt{3}/3$

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